

Long-term Weather Observations Reveal the Impact of Heatwaves on the Yield and Fruit Composition of Cabernet Sauvignon

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Abstract

Background and goals

Warming trends observed globally have fostered research to understand the effects of increasing temperatures on the yield and fruit composition of winegrapes. However, the impact of extreme heat events at the commercial scale remains unsubstantiated. The present study represents the first attempt to combine available long-term spatiotemporal climate data with historical yield and quality metrics to investigate the impact of heatwaves (two or more days with maximum temperature $\geq 38^{\circ}\text{C}$) on commercial winegrape production.

Methods and key findings

Historical weather data (1981 to 2023) for five sites in California were modeled using clustering analysis, separating three clusters: cool seasons, seasons with preveraison heat, and seasons with postveraison heat. Actual or temperature-based phenological stages were used to discriminate heatwave timing. Historical records of fruit composition, yield, and harvest date were then analyzed to determine differences between years experiencing extreme heat events and cool seasons. Yield and harvest date were affected by the intensity and distribution of heat in the growing season. Preveraison heat led to an advancement of harvest and substantial yield losses—on average, 13 days and -30%, respectively. Postveraison heat resulted in lower yield losses (-22%) and even earlier harvest dates (-17 days). Fruit quality parameters most affected by heatwaves were sugar, malic acid, pH, color, and phenolic compounds. Early- versus late-season heat resulted in unique effects on individual quality markers.

Conclusions and significance

By linking climate observations with grape yield and quality metrics at harvest, this study provides measured effects of heatwaves for the grapegrowing industry at a regional scale and points to the need for specific mitigation strategies.

Key words: Cabernet Sauvignon, climate change, grape production, harvest date, heatwaves, yield and fruit composition

Introduction

World grape production is facing the challenges of a climate that is changing rapidly at the global scale (Jones and Webb 2010, van Leeuwen et al. 2024). Among a series of shifts in environmental conditions, viticulture is forced to adapt to increasing worldwide temperatures (Fraga et al. 2012, Garcia-Tejera et al. 2023). Climate analyses have shown increasing trends for most temperature-based or temperature-related indices, including the average and maximum temperature during the growing season, growing degree days (GDD), and other more complex parameters. More frequently, grapegrowing regions must deal with heatwaves, defined as long periods of atypically extreme heat (Parker et al. 2020, Gambetta and Kurtural 2021). Heatwave classification criteria vary in the existing literature, mostly depending on the field of study. In grapes, upper thresholds of 35°C and 38°C have been used to successfully quantify heat exposure levels (Parker et al. 2020, 2024, Gambetta and Kurtural 2021). Metrics such as number of heat days (maximum temperature $[T_{\text{max}}] \geq 35^{\circ}\text{C}$ or 38°C) or heatwaves (≥ 2 or 3 days with $T_{\text{max}} \geq 35^{\circ}\text{C}$ or 38°C) can be used to summarize heat levels in climatic studies. Past and predictive future climate data sets have been leveraged to analyze how heat levels have changed in the past, or to forecast how they will evolve in the future, providing important insights into how climate is changing in grapegrowing regions (Webb et al. 2007, Jones and Webb 2010, Fraga et al. 2016, Parker et al. 2020, Santos et al. 2020, Gambetta and Kurtural 2021). However, climatic data often remain disconnected from horticultural outputs due to a lack of historical data.

To address the question of how vines respond to higher temperatures, decades of research have subjected whole vines, or fruit only, to artificially increased temperatures. In one study, research approaches to study the effect of elevated temperature

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were reviewed, highlighting the benefits and concerns of each method (Bonada and Sadras 2015). Foundational knowledge is derived from exposing vines to increased temperature in controlled environments, where inputs and stress levels can be controlled and unpredictable factors are reduced to a minimum (Kliewer 1977, Edwards et al. 2011, Hewitt et al. 2023). Field experiments more closely resemble commercial production, but it remains difficult to selectively increase temperature due to carryover effects on other parameters (Sadras and Moran 2012, Wu et al. 2019). Another method compared the same vines before and during naturally occurring heat events (Greer and Weedon 2013). These studies fail to describe the unique effects of individual heatwaves (in which severe heat conditions may be repeated over consecutive days), as well as the possible effect of multiple heatwaves per season.

Some studies have directly assessed the effect of heatwaves by simulating temperature increases over several consecutive days: 3 (Gouot et al. 2019), 4 (Greer and Weston 2010), and 7 to 14 days (Lecourieux et al. 2017). In the 3-day study, heatwaves were applied shortly after fruit set (+10°C), at 2 wk after fruit set (+5°C), or as a combination of these application points (Gouot et al. 2019). Yield was not measured, but a transient effect on berry weight was observed only after the first artificial heatwave, and not at harvest. In the 4-day study, Sauvignon blanc-potted vines were subjected to 4-day heatwaves (40/25°C day/night temperature compared to 25/15°C in the control) at flowering, fruit set, veraison, and mid-ripening (Greer and Weston 2010). Yield was affected the most by treatments at flowering and veraison, while berries heated at fruit set were more resilient. Another study pooled data from heating experiments over 7 yr, with an average temperature increase of 2°C at key phenological stages (Sadras et al. 2017). No significant change was found between control and heated vines. Altogether, the effects of elevated temperature on yield, as reported in the literature, remain inconsistent. Impediments to sugar accumulation coupled with decreases in photosynthetic rate were also described when heat occurred at veraison or mid-ripening. Broader detrimental effects of heat on fruit composition and wine quality have also been reported (Mira de Orduña 2010, Sadras et al. 2013, Bonada et al. 2015).

Although some studies have investigated the impact of heat under controlled conditions, there are little data to

substantiate the effects of heatwaves on yield and quality in commercial grape production. One study modeled the impact of heatwaves on yield in European wine regions (Fraga et al. 2020). The authors analyzed yield responses arising from introducing a single heatwave event per growing season, testing the effect of varying heatwave dates and durations (5 to 9 days). Yield reductions were modeled across all regions, with the largest effect being -35% for early August heatwaves. Apart from modeled outcomes, which remain dependent on simulated data, empirical evidence among heat patterns and grape yield and quality has not been described. In this study, we propose that to uncover the degree of change caused by heatwaves, classifications of growing seasons according to heat features can be linked to historical records of grape yield and composition. Hierarchical clustering analysis (HCA) is a statistical technique used to group observations according to similarity. In agrometeorology, HCA (or other techniques of dimensionality reduction, such as principal component analysis) is often employed to classify regions based on their climate (Puga et al. 2022). Our approach consisted of applying HCA to categorize growing seasons according to their weather patterns, with a focus on heat-related aspects. To investigate the irregular nature of hot and cool seasons, which are often alternated over time, categorization of growing seasons was deemed more appropriate than linear regression analysis. This pilot study, conducted on a single variety at the regional scale, provides a foundation for future research to examine other grape varieties and viticultural areas.

Materials and Methods

Vineyard sites and characteristics

Five vineyard sites (S1 to S5) in Northern California were selected for this study on the basis that Cabernet Sauvignon was the predominantly grown grape variety at each site. Site locations are shown in Supplemental Figure 1 and vineyard details are reported in Table 1. Four vineyard sites were located within American Viticultural Areas (AVAs) of the Napa Valley. To introduce variability among the study sites and avoid the classification model becoming too dependent on small regional patterns, the fifth vineyard, located in Sonoma County and comparatively cooler than the other four sites, was added.

Table 1 General description of the five vineyard sites used in this study. AVA, American Viticultural Area.

Site	Location	Geographical classification	Average elevation (m)	No. Cabernet Sauvignon blocks	Average vineyard block size (ha)
S1	38°38'N; -122°27'E	Napa Valley	340	30	4.62
S2	38°20'N; -122°28'E	Moon Mountain AVA	280	56	2.16
S3	38°27'N; -122°18'E	Atlas Peak AVA	455	218	1.18
S4	38°38'N; -122°27'E	Napa Valley	340	18	3.12
S5	38°21'N; -122°16'E	Napa Valley	114	33	2.20

Data collection

This analysis integrated several data sets according to the workflow shown in Figure 1. The data sets consist of two sheets: one containing data and the other a list of variable names and their description. Data collections and curation for each data set are explained below.

Long-term weather observations

Historical weather observations were retrieved from the PRISM Climate Group website (<http://www.prism.oregon-state.edu>), an open-source database that offers gridded long-term climate observations from 1981 to the present, at a spatial resolution of 4 km². Daily data for the five sites are available at Figshare (Database 1; <https://doi.org/10.6084/m9.figshare.28486232>). For each vineyard site, daily weather data from 1 Jan 1981 to 31 Dec 2023 were extrapolated for the grid cell closest to the GPS coordinates reported in Table 1. PRISM data were generated on 7 Feb 2024 and weather records included minimum, average, and maximum temperature (°C); precipitation (mm); and minimum and maximum vapor pressure deficit (kPa).

Phenological stages

Phenological observations recorded in selected blocks were included in the analysis to relate heatwaves to specific growth stages (Database 2; <https://doi.org/10.6084/m9.figshare.28486232>), and historical records for budbreak (BB), flowering (FL), and veraison (VER) were included in

the database. Harvest dates were sourced from the yield data set. When phenological records were missing for one or multiple stages, they were estimated using relationships built between GDD and phenological records collected in the field. Relationships were explored for GDD starting from both 1 Jan and 1 March (Supplemental Figure 2), but GDD_Mar1 was selected due to better performance. GDD averages necessary to reach each phenological stage were calculated by site and year when possible, or by year, depending on data availability. The closest available average was used as an estimate for missing cases, using the long-term GDD average across sites and years only when no data were available. To estimate missing phenological data, PRISM temperature series were leveraged, substituting dates with GDD values associated with BB, FL, or VER.

Harvest, yield, and fruit composition data

All historical records available for Cabernet Sauvignon vineyards within the five sites were included in this analysis (Database 3; <https://doi.org/10.6084/m9.figshare.28486232>). Grafting, planting years, harvest date, and yield were sourced from an internal database. Harvest dates were transformed into day of year (DOY) format to compare across different seasons. If harvest was completed over multiple days, the midpoint date was used. Vineyard yield (tons/ha) was calculated by dividing truckload weights at the winery and vineyard area. Prior to this calculation, data were carefully explored to verify the soundness of the

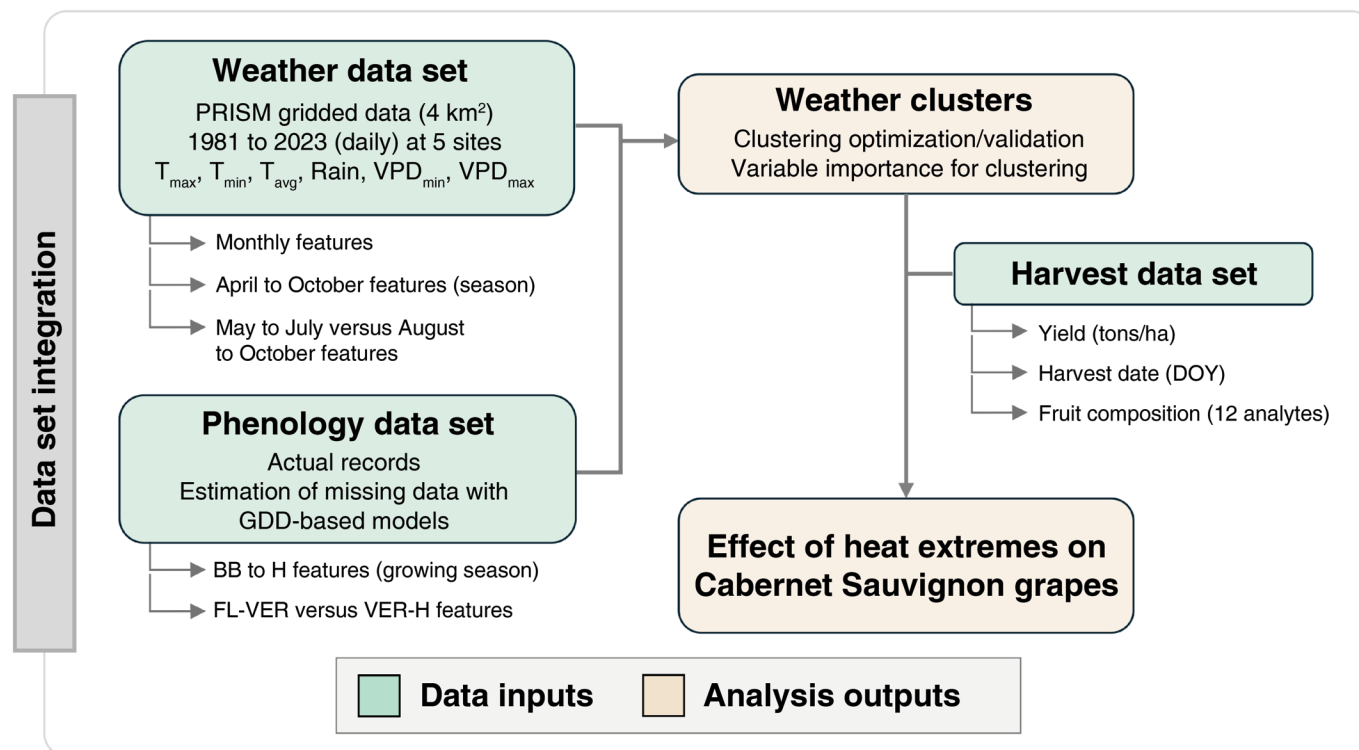


Figure 1 Schematic representation of the workflow of the study, including the collection and integration of different sets of data prior to analysis. T_{max} , maximum temperature; T_{min} , minimum temperature; T_{avg} , average temperature; VPD_{min} , minimum vapor pressure deficit; VPD_{max} , maximum vapor pressure deficit; GDD, growing degree days; BB, budbreak; FL, flowering; VER, veraison; H, harvest; DOY, day of year.

input data. The stability of vineyard size over time was also investigated for each individual vineyard block, and vineyards with unacceptable variation were discarded unless vineyard managers could verify changes over time.

Fruit composition measurements available from 2017 to 2023 (Database 4; <https://doi.org/10.6084/m9.figshare.28486232>) were sourced from the benchmarking program of the winery, based on the identification of key chemical markers to predict Cabernet Sauvignon quality and streamline processing decisions (Cleary et al. 2015). Prior to harvest, one fruit sample (~20 to 50 clusters) was collected from each vineyard and submitted for chemical analysis; samples collected more than a week prior to harvest were discarded. Analytical procedures used to analyze grape samples were standardized across years and described in detail by Previtali et al. (2021). Analytes of interest for this analysis were free (1-octen-3-ol, 3-isobutyl-2-methoxypyrazine [IBMP], and C₆ compounds) and bound (β-damascenone) aroma compounds, color parameters (total anthocyanins), mouthfeel compounds (polymeric tannins and quercetin glycosides), and basic chemistry (total soluble solids [TSS], fruit moisture, pH, malic acid, and yeast assimilable nitrogen [YAN]).

Season classification using HCA

For each combination of year (1981 to 2023) and vineyard site (S1 to S5), long-term weather data were summarized to extract features describing annual weather patterns, with a focus on heat-related indexes. The full list of feature names and descriptions is available (Supplemental Table 1). Heat days were classified as days with $T_{\max} \geq 38^{\circ}\text{C}$, and heatwaves were defined as two or more consecutive days with $T_{\max} \geq 38^{\circ}\text{C}$. Heatwave units (HWU) were calculated as modified GDD by adjusting the minimum threshold to 38°C (Equation 1) across all days of a heatwave.

$$\text{HWU} (^{\circ}\text{C}) = \sum_{\text{Day } n}^{\text{Day } 1} (T_{\max} - 38) \quad (\text{Eq. 1})$$

Summary features were both calculated by month, individually or aggregated, as well as by phenological interval (BB-FL, FL-VER, VER-H). Phenology-based indexes were considered more accurate, but as phenological records are not always available, both chronological and phenological features were included in this analysis to test the level of accuracy when using monthly features. General changes in weather across sites and years were represented with a heatmap after normalizing each variable to a mean of 0 and a standard deviation of 1. Once summarized, scaled features for year and site combinations (e.g., S1_2000) were submitted to HCA to identify groups with similar weather patterns. Clustering performance was screened using three statistical techniques: within-cluster sum of squares, silhouette curve, and gap statistic (Rousseeuw 1987, Tibshirani et al. 2001). In addition, trends were evaluated by cluster for each feature to optimize the meaningfulness of the resulting clusters and maximize the discrimination of seasons for further analysis.

Cluster validation

Cluster stability was assessed using bootstrap random forest (RF; Breiman 2001). The algorithm was used to predict clusters from summarized features, randomly splitting the data set 50:50 for training and testing. Cluster outputs predicted by RF and actual clusters were compared, calculating the percentage of misclassified cases or test error rate (TER, in %). This process was repeated 50 times with bootstrapping, setting the number of trees to 500 and selecting seven random variables for each tree. In addition, variable importance in prediction metrics was retained through the iteration process. The mean decrease accuracy (MDA, in %) across all iterations was calculated for each variable, representing the change in classification accuracy when each variable was omitted.

Season effect on grape yield and composition

To assess the effect of different heat patterns, observations collected for harvest date, yield, and fruit composition were categorized according to the clusters resulting from the analysis above: preveraison heat (PRE-V), postveraison heat (POST-V), and cool season (Cool). Through a process of attribution, weather patterns unique to each cluster were linked to the response variables of interest, following the procedure outlined in Figure 2. Data for a given response variable (e.g., yield) were summarized by clusters at the single vineyard level. Only vineyards with two or more observations for two or more clusters were included in this analysis. Variations due to season type were graphically represented with point and line charts for single vineyards. To generalize across the entirety of the vineyards, density plots were used to show the distribution of observations by cluster. Averages and 95% confidence intervals (CI) by weather cluster were calculated, and relative changes compared to a control (represented by cool seasons) were computed and graphed. The interquartile range (IQR) measure was calculated to compare the degree of dispersion between groups.

Statistical analysis

Data sets were curated in Microsoft Excel (2024) ver. 16.98 and are available at <https://doi.org/10.6084/m9.figshare.28486232>. Statistical analyses were conducted using R (R Core Team 2012) ver. 4.3.3 in RStudio (RStudio Inc.) ver. 2023.12.1, combining available packages (available at <http://www.cran.r-project.org>) and custom-made code. The map of vineyard sites was produced using the R packages 'terra', 'basemaps', and 'rayshader'. AVA shapefiles were retrieved from the University of California Davis Library GitHub repository (Tobias 2022). The heatmap representation of summarized weather data was produced using the 'ComplexHeatmap' package (Gu et al. 2016). Clustering analysis was performed using the package 'factoextra' and the iterative RF was based on functions of the 'randomForest' package. Data visualizations were produced using the 'ggplot2' package (Wickham 2016). Differences in heat features by clusters were tested using one-way analysis of variance, and mean comparison was performed using

Tukey's adjusted post-hoc test with an error rate of $\alpha = 0.05$. Normality of residuals and homoscedasticity assumptions were tested with the Shapiro-Wilk and Levene's tests, respectively. Nonparametric tests were adopted in case of non-normally distributed data, using Kruskal-Wallis to test for factor significance and pairwise Wilcoxon test for mean comparison ($\alpha = 0.05$). Cluster effects on response variables were tested using linear mixed models, with clusters as a fixed effect and individual vineyard blocks as random effect. Mean comparison was performed using Tukey's adjusted post-hoc test with an error rate of $\alpha = 0.05$.

Results

Clustering of growing seasons according to weather features

Summary features ($n = 215$; Supplemental Table 1) extracted from daily weather observations from 1981 to 2023 at five sites (213 combinations in total) are shown (Supplemental Figure 3). Temperature records from PRISM and local weather stations at one of the sites showed highly consistent trends (Supplemental Figure 4). HCA was applied to the scaled data to identify growing seasons with similar weather patterns. The optimal clustering solution was defined by scouting

outputs with an increasing number of clusters (Supplemental Figure 5) and by clustering performance metrics (Supplemental Figure 6). Three clusters maximized cluster distance (Supplemental Figure 6A) and were identified as the best solution according to the silhouette method (Supplemental Figure 6B), so they were adopted for further analysis (Figure 3A). The three-cluster solution was able to differentiate seasons with distinct weather trends by capturing differences in the frequency and temporal distribution of heat events, as visualized in temperature radial charts exemplifying one average observation per cluster (Figure 3B1 to 3B3). The distribution of observations across clusters was Cluster 1 (C1), $n = 21$; Cluster 2 (C2), $n = 70$; and Cluster 3 (C3), $n = 124$. C1 was defined as POST-V, C2 as PRE-V, and C3 as Cool.

Cluster validation

Changes in TER for the bootstrap RF algorithm are shown (Figure 4A). The average TER across 50 iterations was 1.35%. When the number of iterations was gradually increased to 500, the TER remained below 2%, with a maximum TER of 1.63% (Supplemental Figure 7). The most important variables for RF classification, selected as those with MDA $\geq 1\%$, are shown (Figure 4B). Highly predictive features were all heat-related, including the number of

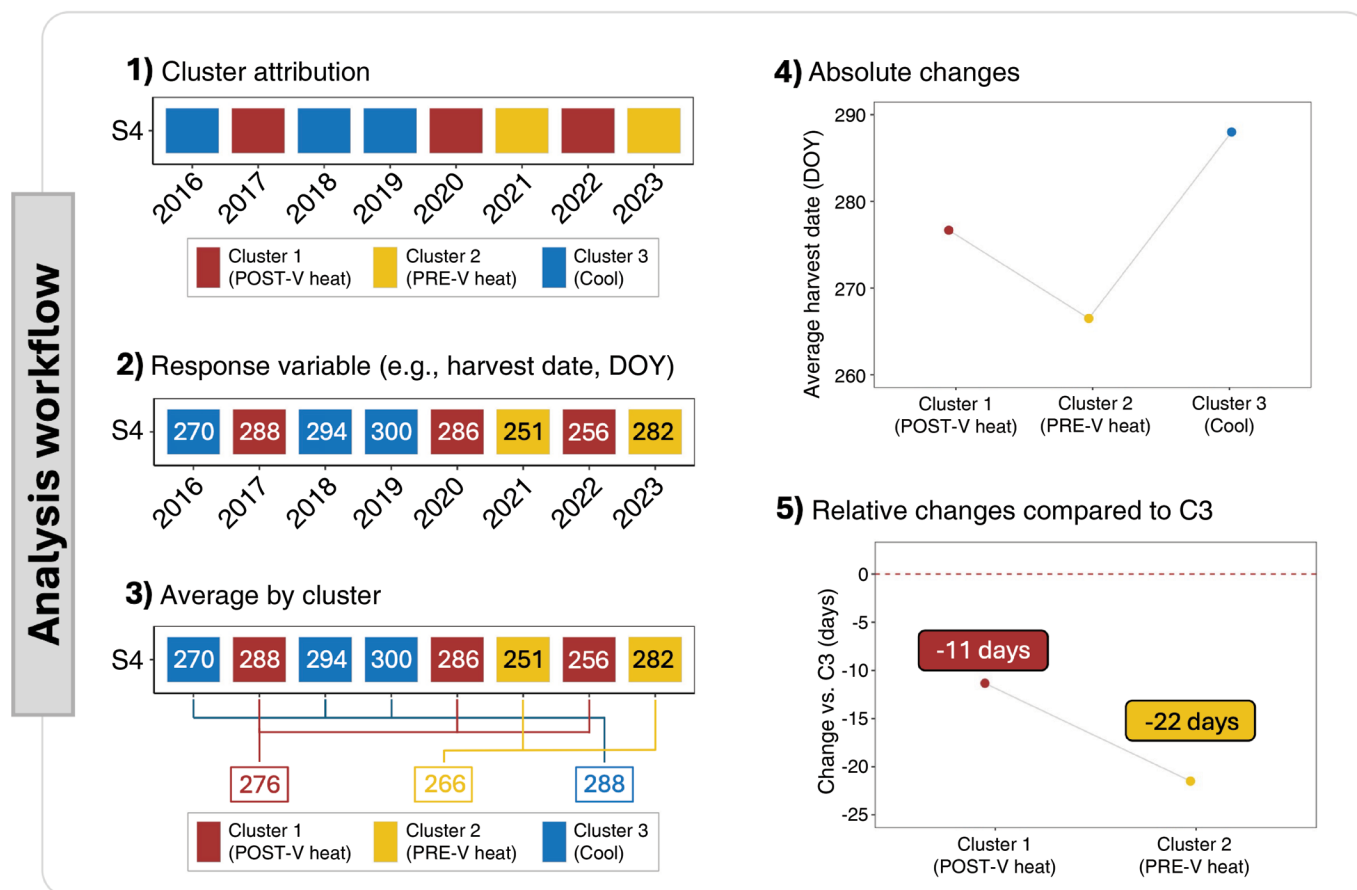


Figure 2 Schematic step-by-step representation of the procedure followed to link weather-based clusters to yield, harvest date, and fruit composition. Harvest date records for one vineyard at site 4 (S4) from 2016 to 2023 are shown as an example. POST-V, postveraison heat; PRE-V, preveraison heat; Cool, cool season; DOY, day of year; C3, Cluster 3.

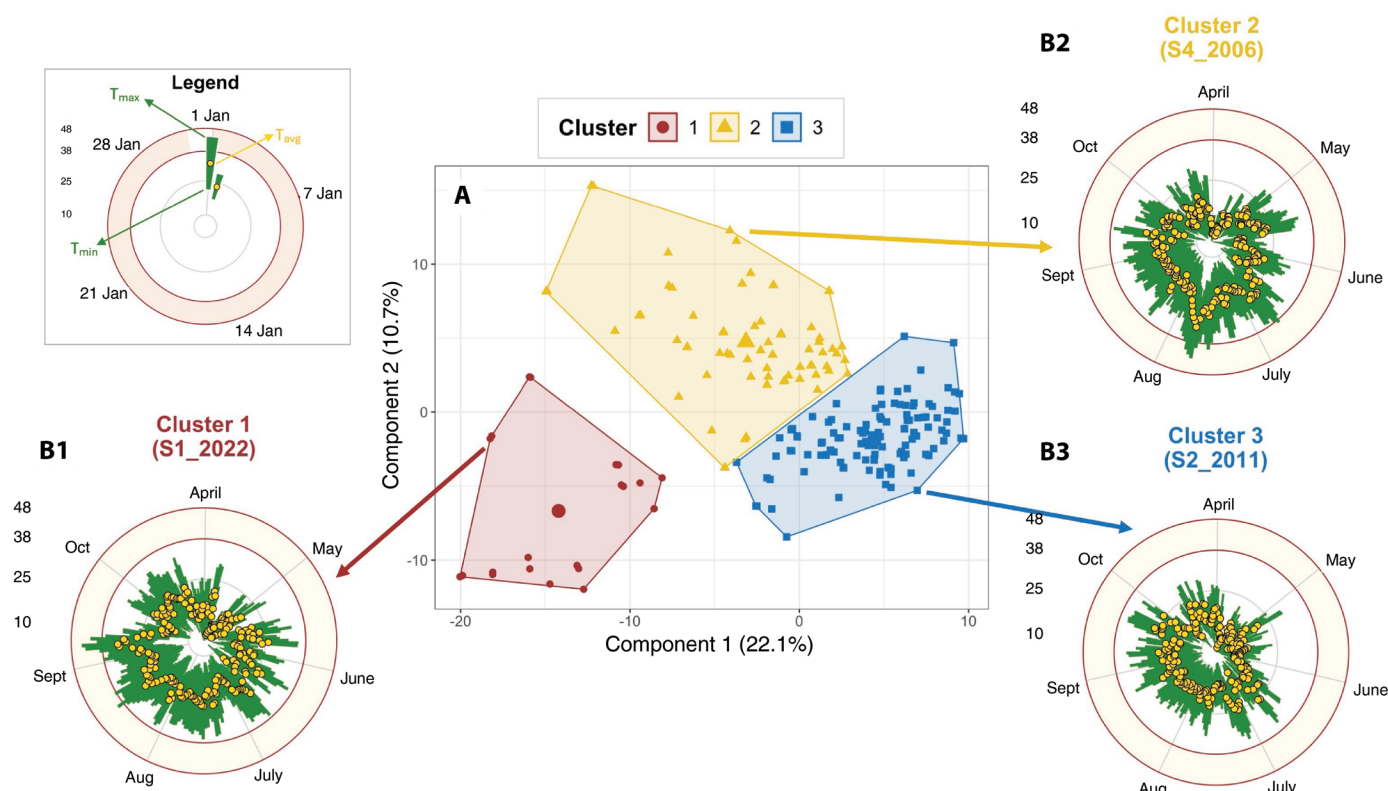


Figure 3 Hierarchical clustering analysis (HCA) of historical weather observations (1981 to 2023) at five vineyard sites in California (S1 to S5). **A**) Clusters ($n = 3$) identified using HCA, shown using a principal component analysis score plot. **B1** to **B3**) Temperature radials (April to October) showing unique patterns associated with each cluster. In each graph, temperatures are shown for a single site and year for exemplification purposes. Green bars represent the range between daily minimum and maximum temperatures ($T_{\min}$ and $T_{\max}$, respectively; $^{\circ}\text{C}$); yellow points indicate daily average temperatures (T_{avg}). The shaded yellow band marks temperatures above the critical threshold of 38°C .

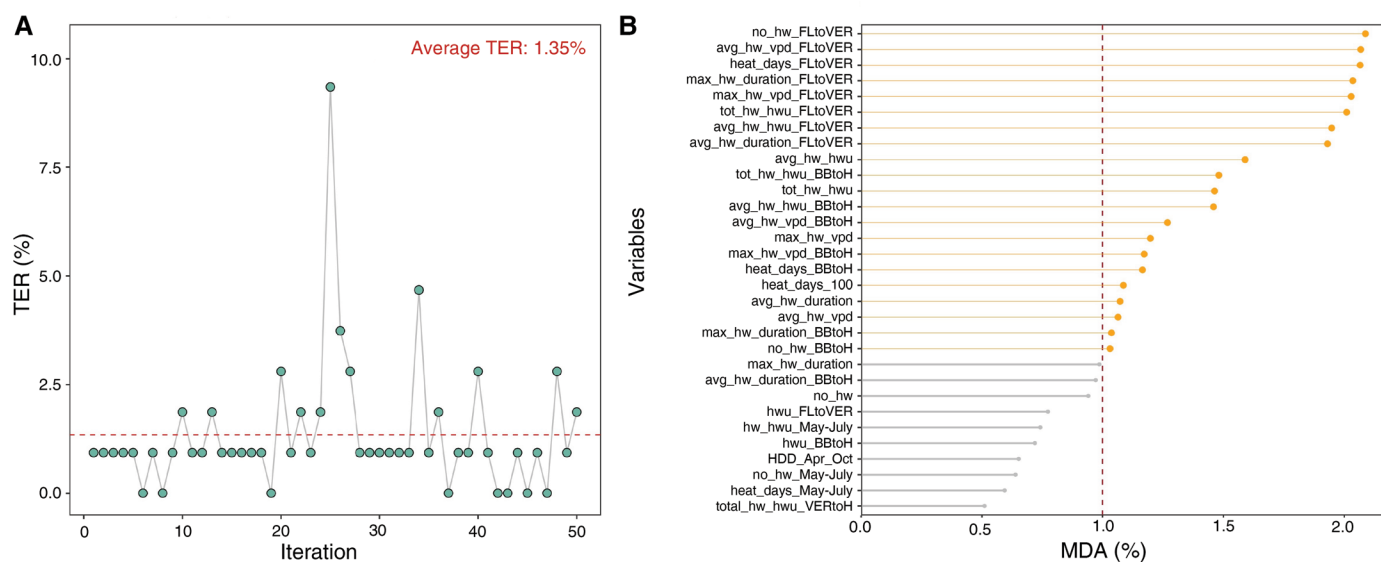


Figure 4 Cluster validation using bootstrap random forest classification. **A**) Test error rate (TER) representing the percentage of misclassified observations at each iteration. Green points represent the error rate at each iteration. The red dotted line represents the average TER; the value is reported in the label at the top of the graph. **B**) Variables with the highest contribution for cluster classification, represented by the mean decrease accuracy (MDA). The full list of weather features and their description is available in Supplemental Table 1. Only variables with $\text{MDA} > 0.5\%$ are reported in this chart. Variables with $\text{MDA} > 1\%$ are marked in orange and represent the most important features driving cluster classification. FL, flowering; VER, veraison; HWU, heatwave units; BB, budbreak; H, harvest.

heatwaves and heat days, as well as heatwave characteristics such as duration and VPD of heat events. The first eight most contributing variables, all features based on phenological boundaries, referred to the preveraison stage (FL to VER); these were followed by similar features but calculated for the entire season using chronological boundaries (April to October) or actual BB and harvest records (BB to H).

Cluster characteristics

Trends by clusters were explored for all features (Supplemental Figures 8 to 13), however, the largest differentiation was observed for $T_{\max}$, heat days, and heatwaves (Figure 5). The average $T_{\max}$ in the growing season was just below the critical threshold of 38°C for C3 (37.9°C; Figure 5A), while it was significantly higher in both C1 (42.2°C) and C2 (40.1°C). Similarly, there were differences in the monthly distributions of $T_{\max}$ by cluster. In C1, $T_{\max}$ distribution was increased in August and even more in September, while the distribution of C2 showed an increase in July. Summarized values of $T_{\max}$ by month confirmed these trends. When data were summarized by phenological intervals, average $T_{\max}$ in C3 was below 38°C both pre- (FL to VER, 37.4°C) and postveraison (VER to H, 37.0°C). Compared to C3, $T_{\max}$ was higher preveraison but unchanged postveraison in C2 (40.1°C and 37.5°C, respectively), while the opposite trend was observed for C1 (38.8°C preveraison; 42.0°C postveraison). Results obtained by summarizing $T_{\max}$ by phenological stage and by groups of months were very similar (Figure 5A). Trends by cluster for features involving the number of heat days are shown (Figure 5B). The total number of heat days in the growing season was highest in C1 (5.20 days on average), followed by C2 (2.94 days), and in C3, it was close to zero (0.17 days). Heat days by month in C3 were close to zero across the board, while C2 had a higher number of heat days in June and July (0.54 and 1.82 days on average, respectively), or preveraison, when using phenological stages as boundaries (2.74 days from FL to VER). On the contrary, C1 reached the highest number of heat days in August (2.19 days) and September (2.57 days) and in the postveraison stage (5.24 days). Trends for heatwave features are shown (Figure 5C) and matched results observed for $T_{\max}$ and heat days. On average, the number of heatwaves per season in C3 was negligible (0.1), while there was on average at least one preveraison heatwave in C2 and close to two heatwaves mostly occurring postveraison in C1. Heatwave duration and maximum VPD reflected cluster differences reported for other features: 3.2 days on average and a maximum VPD of 7.2 kPa in C1; 2.5 days and 6.1 kPa in C2; and no heatwaves (0.2 days) in C3.

Effect of heat extremes on yield and harvest date

Exploratory analysis of the yield data set is available (Supplemental Figure 14), and harvest dates by vineyard block as

a function of season clusters are shown (Figure 6A and 6B). Harvest occurred on average 17 and 13 days earlier in hot years (C1 [DOY 291] and C2 [DOY 274], respectively; $p < 0.0001$ for both C1 and C2) when compared to cool years (C3 [used as a reference]). Harvest dates of C1 and C2 were also significantly different ($p = 0.001$). The dispersion of harvest dates was greater in C1 (IQR = 9.8 days) and C2 (IQR = 15.3 days) compared to C3 (IQR = 6.5 days).

Regarding the effect of the clusters on vineyard yield, C3 had the largest yield (7.0 tons/ha) and there was a significant reduction in C1 (5.5 tons/ha, $p < 0.0001$), with an even larger reduction in C2 (5.0 tons/ha, $p < 0.0001$) (Figure 6C). The difference in yield between C1 and C2 was significant at $p = 0.010$. Based on a summary of percent change values by block compared to C3, yield was decreased by 22% and 30% on average in C1 and C2, respectively (Figure 6D). Yield data were also analyzed by grouping vineyards by vine age (Supplemental Figure 15): young (3 to 5 yr) and mature (>5 yr). There was a significant interactive effect between season clusters and vineyard age ($p = 0.037$). Cluster means were all significantly different from each other ($p = 0.029$ or lower) in mature vineyards and reflected differences reported for pooled data: C1 (5.4 tons/ha), C2 (4.9 tons/ha), C3 (7.1 tons/ha). In young vineyards, yield was unchanged in C1 and C2 (5.4 tons/ha in both groups) and both heat-related groups had significantly lower yield when compared to C3 (6.5 tons/ha, $p = 0.001$ or lower).

Effect of heat extremes on fruit composition at harvest

Exploratory analysis of the fruit composition data set is reported (Supplemental Figure 16). Cluster effects on grape analytes are shown (Figure 7) and detailed statistics for each analyte are reported (Supplemental Table 2). All grape analytes except C6 compounds ($p = 0.371$) and berry moisture ($p = 0.299$) were significantly affected by season clusters ($p \leq 0.001$). 1-Octen-3-ol levels were highest in C1 (50.2 µg/L), followed by C2 (27.8 µg/L), while the lowest concentrations were associated with C3 (17.1 µg/L). Polymeric tannins followed the same trend, displaying decreasing levels from C1 (3856 mg/L) to C2 (3447 mg/L) and C3 (2979 mg/L). Malic acid, TSS, and pH were unchanged in C2 and C3 (1338 g/L, 26.0 Brix, and 3.60, respectively) but increased in C1 (1996 mg/L, 26.4 Brix, and 3.65, respectively). Contrarily, quercetin glycoside levels were equal in C1 and C3 (118 mg/L) and significantly higher in C2 (146 mg/L). β-Damascenone was decreased to the same extent in C1 and C2 (53 µg/L) compared to C3 (57 µg/L). YAN was also negatively affected by heat events, but the effect versus C3 (124 mg/L) was more pronounced in C2 (66 mg/L) than in C1 (94 mg/L). IBMP and total anthocyanins decreased in C1 (2.71 ng/L and 1.50 mg/g, respectively) but increased in C2 (5.7 ng/L and 1.85 mg/g, respectively), and cool seasons of C3 had intermediate values (4.1 ng/L and 1.75 mg/g, respectively).

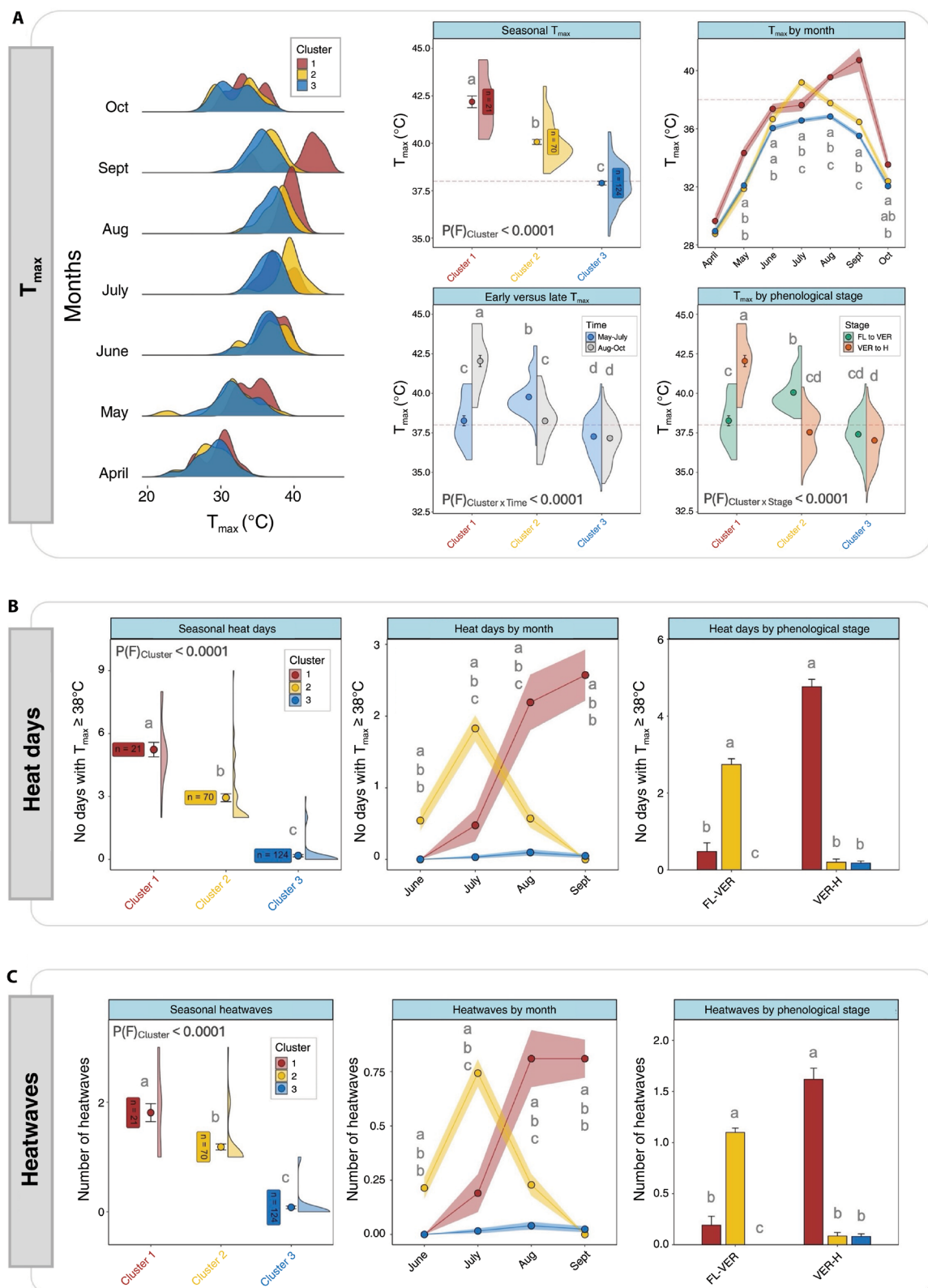


Figure 5 Intensity and distribution of heat events for three clusters identified using hierarchical clustering analysis. **A**) Maximum temperature (T_{max} ; °C): distribution by month for each cluster, season (April to October), monthly T_{max} by cluster, and comparison of T_{max} distributions by phenological stage and month intervals to approximate pre- (May to July) and postveraison (August to October). **B**) Number of heat days ($T_{max} \geq 38^{\circ}\text{C}$): total heat days in the growing season (April to October), by month (June to September), and by phenological stage. **C**) Number of heatwaves (two or more days with $T_{max} \geq 38^{\circ}\text{C}$) in the growing season (April to October), by month (June to September), and by phenological stage. Density curves represent distributions by cluster. Points and error bars or ribbons represent means $\pm$ 95% confidence intervals. Letters denote significant differences ($p \leq 0.05$) according to one- or two-way analysis of variance and Tukey's adjusted post-hoc test in case of normal data, or Kruskal-Wallis followed by pairwise Wilcoxon test for non-normally distributed data. FL, flowering; VER, veraison; H, harvest.

Discussion

To support the objective of this study, we propose a workflow to quantify the effect of heat extremes (i.e., heatwaves) at the commercial scale by linking long-term weather observations (1981 to 2023) to historical records of grape yield and fruit composition of Cabernet Sauvignon at harvest. For the first time, heatwave effects on key grape production metrics were determined, based on the integration of large phenological, weather, and grape production data sets (Figure 1).

Summary features extracted from weather series (Supplemental Table 1) were used to group similar growing seasons via clustering analysis. In previous research, changes in given parameters were assessed over time

to identify positive or negative effects over the period investigated (e.g., Gambetta and Kurtural 2021, Faralli et al. 2024). To address the residual year-to-year variability observed in the historical data analyzed in these studies, our approach was to classify years according to their weather patterns. Of the 220 weather-based features created, several were aimed at quantifying levels of heat, such as the number of heat days and heatwaves (base temperature of 38°C); PRISM weather observations were used. Others have addressed the benefits and limitations of spatially interpolated data (Schneider and Ford 2013, Van Wart et al. 2013, Mourtzinis et al. 2017). A certain degree of inaccuracy was accepted in this analysis (see comparison with local weather station in Supplemental Figure 4), as PRISM offered historical weather

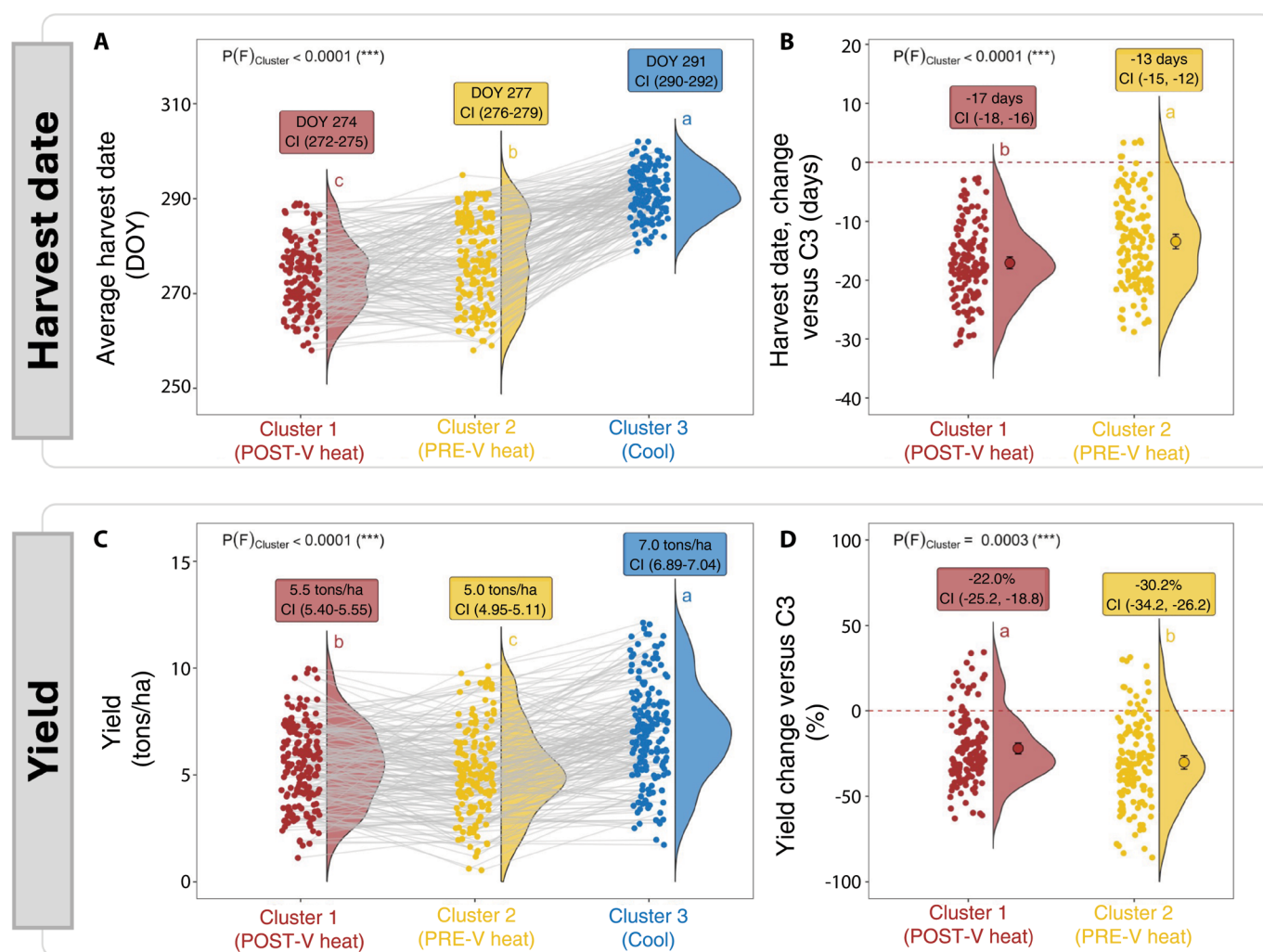


Figure 6 Effect of heat extremes on Cabernet Sauvignon grape harvest date (**A** to **B**) and yield (**C** to **D**). Differences in heat patterns during the growing season are represented by three clusters: Cluster 1 (POST-V, postveraison heat), Cluster 2 (PRE-V, preveraison heat), and Cluster 3 (Cool, cool seasons). In **A** and **C**, points represent averages for individual vineyard blocks and density plots show distributions for each cluster. Gray lines connect averages belonging to an individual vineyard block across cluster types. In **B** and **D**, small points represent changes for C1 and C2 compared to C3, and density plots show distributions by cluster. The large point represents the mean effect, and error bars represent 95% confidence intervals (CI). The red dotted line represents no effect. Averages and CIs by cluster are annotated in text boxes above the data. Across the board, p values denote statistical significance of cluster effects according to a linear mixed model (fixed effect: cluster; random effect: vineyard block ID) and letters denote differences across means by clusters according to Tukey's adjusted post-hoc test ($p \leq 0.05$). *** indicates statistically significant differences at $p < 0.001$. DOY, day of year.

series deemed suitable to characterize year-to-year differences. HCA results (Figure 3) discriminated three groups of seasons with distinct heat intensity and temporal distribution. Variables with the largest contribution to this separation were directly related to heat indices, defined by phenological stage or time of year (Figure 4B). C3 was dominated by the lowest values for all heat-related features, with $T_{\max}$ well below the critical threshold of 38°C and zero heat days and heatwaves per season (Figure 5). Seasons of the remaining two clusters displayed much higher heat levels than C3, although they occurred at different times of the growing season. This indicates that the clustering algorithm was able to discriminate seasons according to both the intensity and timing of heat events. Cluster 1 described seasons with the most extreme heat (Figure 5), namely, the highest $T_{\max}$ and number of heat days and heatwaves, and the lowest rainfall levels (Supplemental Figure 13). In C1, heat extremes were predominantly distributed in the postveraison period (VER to H; Figure 5A to 5C). Levels of heat in C2 were still higher than the critical threshold, but heat spikes occurred preveraison (FL to VER). The robustness of this clustering solution was validated using the RF algorithm (Figure 4). Even under strict conditions such as bootstrapping and dividing the data set 50:50 for training and testing, the clustering was extremely stable, with an average error rate below

2%. Correct cluster classification was highly dependent on heat-related features, especially those calculated using phenological stages (Figure 4B). We observed highly similar trends for heat-related variables when expressed using chronological and phenological intervals (Figure 5A). These results are site- and variety-dependent, and the alignment between aggregated months and phenology may disappear or require recalibration when a similar approach is applied elsewhere. Given the link between heatwave timing and production performance of Cabernet Sauvignon reported in our analysis, phenological data should be preferred when available to align heat events with specific stages of grape reproductive development.

The objective of this study was to understand how heatwaves, defined as events where heat intensity and duration are well above plant functional and productive optimums, affect vineyard production parameters, namely harvest date, vineyard yield, and fruit composition. This was achieved through a process of attribution, summarizing each response variable by cluster, representing season types (Figure 2). Data were summarized for individual vineyards then all observations were pooled into large distribution curves for cluster comparisons (Figures 6 and 7). Given the large number of historical records, heatwave effects were visualized as clear shifts in the distribution of harvest

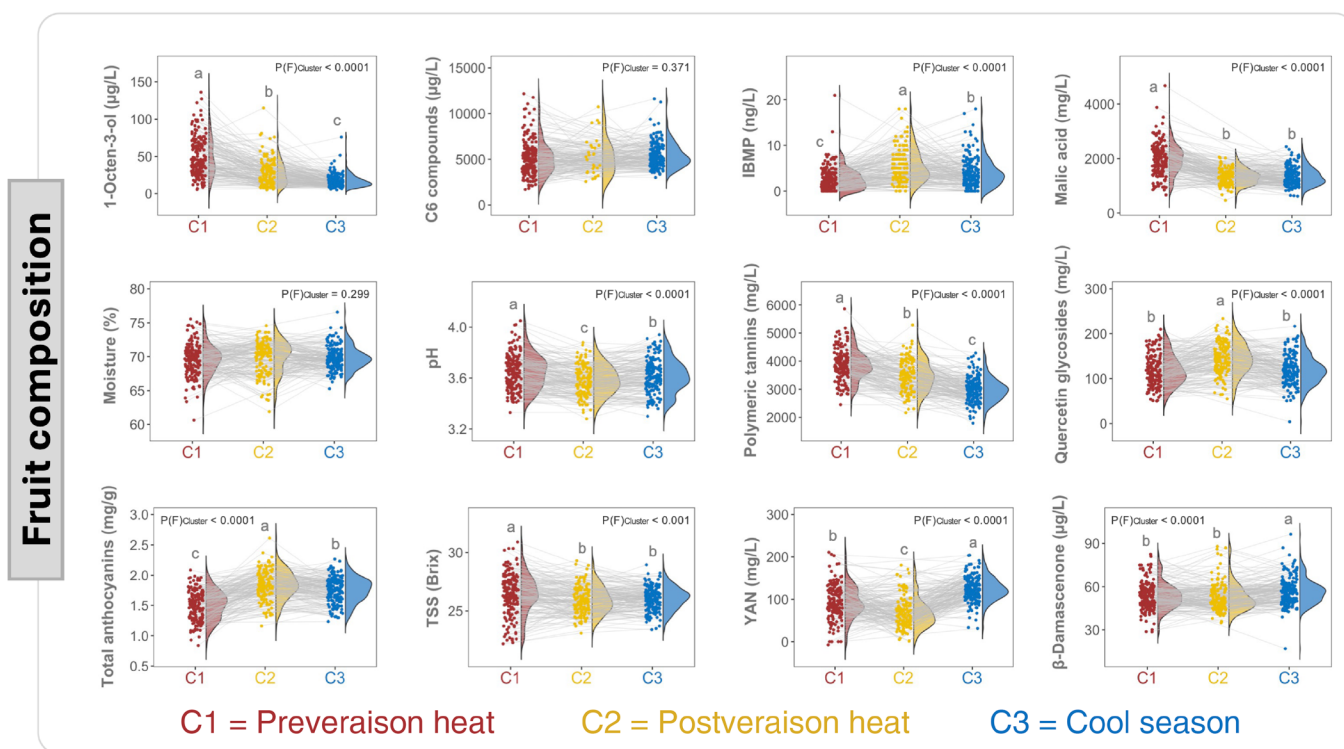


Figure 7 Effect of heat extremes on the concentration of key grape analytes in Cabernet Sauvignon fruit. Each analyte is reported in a separate panel and respective units are reported on each chart. Differences in heat patterns during the growing season are represented by three clusters: Cluster 1 (C1), Cluster 2 (C2), and Cluster 3 (C3). Points represent averages for individual vineyard blocks and density plots show distributions for each cluster. Across cluster types, gray lines connect averages belonging to an individual vineyard block. Across the board, p values denote statistical significance of cluster effects according to a linear mixed model (fixed effect: cluster; random effect: vineyard block ID) and letters denote differences across means by clusters according to Tukey's adjusted post-hoc test ($p \leq 0.05$). IBMP, 3-isobutyl-2-methoxypyrazine; TSS, total soluble solids; YAN, yeast assimilable nitrogen.

date, yield, and several fruit quality markers. Harvest date was anticipated in seasons with both PRE-V (C2) and POST-V (C1), on average 17 and 13 days, respectively (Figure 6A). Earlier harvest under warmer conditions is explained by advancing grapevine phenology from higher temperatures (Webb et al. 2007, Fraga et al. 2016, Ramos et al. 2018). Additionally, heat stress and water deficit are known to accelerate berry dehydration (Bonada et al. 2013, Xiao et al. 2018), which increases sugar concentration. Earlier harvest dates in C1 may also reflect attempts to avoid late-season heatwaves (Figure 5C) by harvesting early, especially in highly susceptible vineyards or to attain specific wine styles. Another effect observed in heatwave seasons (C1 and C2) was a wider harvest window compared to cool seasons of the C3 cluster, as evident in more variable distributions of harvest date points (C1 and C2; Figure 6A). This may be due to vineyard blocks displaying different sensitivities to heat, or specific management practices applied to reduce heat effects on grapevines, which were not included or considered in this analysis (Valentini et al. 2021, 2024, Marigliano et al. 2022, Previtali et al. 2023, Wilson et al. 2024).

Although yield differences between heatwave years (C1, C2) and cool seasons (C3) (Figure 6C) were not as striking as harvest date trends, these results were still highly significant and of value, considering that they were drawn from commercial data. A larger variation in commercial data over that from conventional research trials is expected, as well-defined experimental conditions are used to evaluate heat effects in conventional trials. However, the data approach used in this study relies on a much larger sample size than in research-scale trials, improving the confidence of estimated effects. Notably, negative shifts in yield distributions of C1 and C2 (indicating average yield losses of 30% under PRE-V and 22% with late-season heatwaves) confirm that commercially implemented mitigative actions are often overridden by heat extremes such as those referenced above. These values are within the range predicted in heatwave simulations, where yield drops up to 35% have been modeled in response to a single heatwave per season (5 to 9 days, different time of the year depending on the region) in European wine regions (Fraga et al. 2020). These results also support previous evidence that grapes are more susceptible to heat preveraison, here quantified as an additional average yield drop of ~10%. These findings of increased grape susceptibility to PRE-V are in general agreement with a prior study which described transient effects on Shiraz berry weight after 3-day heatwaves postflowering (Gouot et al. 2019). In a different study, 4-day heatwaves were simulated at different stages of growth on potted Sauvignon blanc (Greer and Weston 2010), and yield was reduced the most at fruit set (–62%) and mid-ripening (–53%), and less at veraison (–31%). Effects of elevated temperature on cell division and expansion processes during the first stage of berry growth are well documented (Kliewer 1977). Impediments to berry growth may be mediated by water deficit prior to veraison (Ojeda et al. 2001, Keller et al. 2008), or the effects could be combined, depending on the exact timing of the heat with

respect to phenological development in each year. Other factors could be included in this analysis to further account for within-cluster variability. For example, we repeated the same analysis, splitting vineyards by age (Supplemental Figure 15), and found that yield of young vines was equally affected in C1 and C2 compared to C3, while in mature vines C2 displayed a larger yield reduction compared to C3. Similarly, application of cultural practices (e.g., canopy management, fruit thinning, irrigation, floor management) to align vine response to production targets could improve this analysis, though such practices are often not accessible and difficult to model quantitatively.

The same analysis presented for yield and harvest date was repeated on grape composition data that were obtained by analyzing key grape quality markers in commercial samples prior to harvest for over a decade. The grape analytes were selected due to high correlations with wine quality scores in Cabernet Sauvignon (Cleary et al. 2015). Heat patterns profoundly affected grape composition, as shown by the large number of analytes that were significantly different between clusters (Figure 7). Pre- and postveraison heatwaves differentially affected the aroma profile of grapes, but the effect was negative in both cases. POST-V (C1) events increased the off-flavor 1-octen-3-ol and decreased fruity β -damascenone but also reduced the concentration of IBMP, which results in undesirable green aromas in wine. PRE-V (C2) did not change 1-octen-3-ol compared to Cool, while β -damascenone declined and IBMP levels increased, moving the green aroma-to-fruit aroma ratio negatively toward green aromas. The decrease of β -damascenone in heated grapes was not reported before, however, changes in the aroma profile from “fresh” to “cooked” fruit have been reported in wines made from heat-treated vines (Bonada et al. 2013, 2015). This change in aroma profile may be explained by norisoprenoid reductions and the downregulation of genes that are involved in the biosynthesis of terpenes and norisoprenoids, as previously described in heat-treated Cabernet Sauvignon (Lecourieux et al. 2017). The observed reduction of IBMP in seasons with POST-V (C1) is in agreement with prior findings in which IBMP decreased at bunch closure in Cabernet Sauvignon under higher temperatures (+1.5°C on average) that were obtained using open-top greenhouses, while differences disappeared at harvest (Wu et al. 2019). IBMP generally responds negatively to heat and sunlight exposure (Falcão et al. 2007, Koch et al. 2010) and heat treatments repress the transcription of the *VviOMT3* gene, a process that is essential for the biosynthesis of IBMP (Lecourieux et al. 2017). The increase observed for seasons with PRE-V (C2) may represent a negligible effect (1 ng/L), especially considering the low perception threshold (7 ng/L; Sidhu et al. 2015).

Across several premium Cabernet Sauvignon vineyards, the concentration of grape phenolic compounds at harvest was greatly affected in heatwave years and the timing of heat affected specific phenolic measures (Figure 7). Our data approach returned significant increases in anthocyanin concentrations during early heatwave years (C2), as

well as significant decreases during late heatwave years (C1). Consistent with our results, preveraison heatwaves did not affect anthocyanins (Gouot et al. 2019); the expression of anthocyanin-related genes was not affected when green berries were submitted to heat, but it was reduced with heat after veraison (Lecourieux et al. 2017). When temperature was increased throughout the growing season using open-top chambers (Sadras and Moran 2012, Wu et al. 2019), anthocyanins were also found to be negatively affected as the fruit was exposed to heat until maturity.

For grape skin tannins, the present analysis returned positive effects of pre- (C2) and postveraison (C1) heatwaves on polymeric tannins, positively affecting wine quality (Cleary et al. 2015), and increases in quercetin glycosides resulted from preveraison heatwaves only (C2). In an earlier study investigating the effect of early heatwaves on grape skin tannins, it was found that the effect of the heat events (with respect to tannin biosynthesis dynamics) depended on the timing of heat (Gouot et al. 2019). The differential response to heat based on phenological stages in this study is supported by other work that showed that the modulation of genes in biosynthesis of phenolic compounds varies depending on the time of the heat treatment (Lecourieux et al. 2017). In contrast, Wu et al. (2019) reported no changes in tannin profile from increasing the temperature by 1.5°C. Overall, the findings from the present study demonstrate that elevated temperature conditions enhance grape tannin concentration, likely driven by increased biosynthesis and potentially amplified by dehydration effects.

Grape basic chemistry at harvest seemed to be solely affected by postveraison heatwaves (C1), resulting in higher TSS, pH, and malic acid (Figure 7). One likely reason for this finding is that after fruit is exposed to heat preveraison (C2), grape basic chemistry may still be improved in the remaining part of the season (Lecourieux et al. 2017). According to the cluster characterization (Figure 5), weather patterns for C2 resembled C3 more than C1 during the ripening stage, possibly indicating conditions that are close to optimum for sugar accumulation and the retention of malic acid. Such compensation cannot occur when heatwaves occur postveraison (C1), particularly when they are very close to harvest. The higher TSS levels observed for C1 could be attributed to the promotion of photosynthetic activity under moderate temperature increases (Greer and Weedon 2012), or to the concentration effects driven by berry dehydration under heat and water stress (Xiao et al. 2018). According to berry moisture data, no significant dehydration effect was observed between clusters (Figure 7). The absence of differences could be attributed to data variability and the fact that large changes in berry weight cause very little shifts in berry moisture (15 to 20% change in berry weight leads to 1 to 2% change in berry moisture [data from another study and not shown]). In addition, some level of berry shriveling may be desired to achieve specific wine style targets (Antalick et al. 2021). In the absence of berry weight data (which were not available for this analysis), it is not possible to address whether higher TSS resulted

from enhanced active sugar accumulation or from higher berry dehydration. One trend that supports the hypothesis of berry dehydration is the increased level of malic acid under POST-V (C1) compared to PRE-V (C2) and Cool (C3). Malic acid is synthesized in the fruit preveraison and degraded during the ripening stage via respiration, a process that is exacerbated by higher temperatures (Sweetman et al. 2014, Rienth et al. 2016).

In summary, this study presents robust evidence that the commercial production of Cabernet Sauvignon is largely affected by heatwaves. By analyzing data from a large pool of vineyards over multiple decades, it was revealed that both pre- and postveraison heatwaves substantially reduced yield and anticipated harvest and altered the concentration of key quality markers at harvest. Further research is necessary to extend this work to other growing regions and varieties, which may lead to identification of genotypes that are less susceptible to heatwaves, or to identification of improved genotype–environment combinations that are better suited to warmer and drier conditions.

Conclusion

This study evaluated the effect of heat extremes in commercial vineyards. Differences in heat intensity and distribution, uncovered by clustering growing seasons according to their weather patterns, were linked to yield losses and diminished fruit composition in Cabernet Sauvignon grapes at harvest. These findings report quantitative data on the commercial impact of heatwaves, an increasingly frequent challenge for viticulturists, and strengthen results from previous field experiments and modeling approaches. Although we focused on a single variety at a small regional scale, the results of this study open opportunities to identify the impact of heat extremes at a larger regional and varietal-specific scale. This study also presents foundational evidence that supports the implementation of strategies that will help to maintain appropriate grape yield and quality in vineyards that are increasingly exposed to more frequent and severe heat events.

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Supplemental Data

The following supplemental materials are available for this article at ajevonline.org:

Supplemental Table 1 List of summary features extracted from PRISM long-term weather observations. GDD, growing degree day; $T_{\max}$, maximum temperature; HWU, heatwave units; DOY, day of year; T_{mean} , mean temperature; $T_{\min}$, minimum temperature; VPD, vapor pressure deficit.

Supplemental Table 2 Statistical outputs of harvest date, yield, and fruit composition parameters by cluster (C1, cluster 1, postveraison heat; C2, cluster 2, preveraison heat; C3, cluster 3, cool season). *P* values of cluster effects were computed using linear mixed models

(fixed effect: cluster; random effect: vineyard block ID). Data by cluster are presented as means $\pm$ 95% confidence intervals and different letters denote significant differences between means according to Tukey's adjusted post-hoc test at $p \leq 0.05$. Significant p values (i.e., ≤ 0.05) are marked in bold formatting. DOY, day of year; TSS, total soluble solids; YAN, yeast assimilable nitrogen; IBMP, 3-isobutyl-2-methoxypyrazine.

Supplemental Figure 1 Map of vineyard site (S1 to S5) locations and closest American Viticultural Areas (AVA).

Supplemental Figure 2 Relationships between observed phenological stages (BB, budbreak; FL, flowering; VER, veraison; H, harvest) and cumulative growing degree day (GDD) calculated from **A**) 1 January or **B**) 1 March. Cumulative GDD were calculated using PRISM daily data.

Supplemental Figure 3 Heatmap of scaled historical weather data submitted to clustering analysis. Columns represent combinations of a vineyard site (S1 to S5) and season (1981 to 2023), rows show normalized values (mean = 0, standard deviation = 1) for each weather feature extracted from historical series. Dendrograms group rows and columns using hierarchical clustering analysis. Gray boxes represent categories of features (generic weather and heat-related) and specify whether growth stages were based on calendar days or phenological stages.

Supplemental Figure 4 Comparison between daily temperature values from PRISM (gridded climate data) and a local weather station at one of the study sites (April to October): average (T_{avg} ; **A** and **B**), maximum (T_{max} ; **C** and **D**), and minimum temperature (T_{min} ; **E** and **F**). In **A**, **C**, and **D**, lines connect daily temperature values and colors differentiate between data sources. **B**, **D**, and **F** show daily differences between PRISM and local station data.

Supplemental Figure 5 Hierarchical clustering analysis outputs by varying number of clusters from 2 to 10 clusters. The three-cluster solution was selected as the optimal number of clusters for further analysis.

Supplemental Figure 6 Hierarchical clustering analysis (HCA) optimization with clustering performance metrics. Three different indexes were assessed in response to varying number of clusters from 1 to 10: **A**) within sum of squares (WSS); **B**) silhouette width; **C**) gap statistic.

Supplemental Figure 7 Cluster validation using iterative bootstrap random forest classification with three clusters. Each panel represents the test error rate (TER) for a given number of iterations, from 50 to 500. Green points represent TER values for each iteration. The red dashed line represents the average TER across all iterations, with the value annotated in red at the top of each graph.

Supplemental Figure 8 Growing degree day (GDD) features summarized by weather cluster. **A**) Distribution of total GDD (April to October) by cluster; **B**) monthly distribution of GDD averages by cluster; **C**) GDD by cluster early and late in the growing season; and **D**) monthly distribution of GDD values by cluster. Colors differentiate between clusters, indicating seasons with different heat levels: Cluster 1, postveraison heat; Cluster 2, preveraison heat; Cluster 3, cool seasons. When present, points and error bars represent means $\pm$ standard error by cluster.

Supplemental Figure 9 Heatwave unit (HWU) features summarized by weather cluster. HWU was calculated as the cumulative deviation of T_{max} -38°C , considering positive deviations only. **A**) Distribution of total HWU (April to October) by cluster; **B**) monthly distribution of HWU by cluster; **C**) HWU by cluster early (May to July) and late (August to October) in the growing season; **D**) summary of HWU by phenological stage (BB, budbreak; FL, flowering; VER, veraison; H, harvest); and **E**) monthly distribution of HWU values by cluster from May to October. Colors differentiate between clusters, indicating seasons with different heat levels: Cluster 1, postveraison; Cluster 2, preveraison heat; Cluster 3, cool seasons. When present, points and error bars represent means $\pm$ standard error (SE) by cluster. Ribbons represent SE values in line charts.

Supplemental Figure 10 Average temperature (T_{avg}) features summarized by weather cluster. **A**) Distribution of seasonal T_{avg} values by cluster (April to October); **B**) monthly distribution of T_{avg} values by cluster; **C**) T_{avg} values by cluster early (May to July) and late (August to October) in the growing season; and **D**) monthly distribution of T_{avg} values by cluster. Colors differentiate between clusters, indicating seasons with different heat levels: Cluster 1, postveraison heat; Cluster 2, preveraison heat; Cluster 3, cool seasons. When present, points and error bars represent means $\pm$ standard error (SE) by cluster. Ribbons represent SE values in line charts.

Supplemental Figure 11 Minimum temperature (T_{min}) features summarized by weather cluster. **A**) Distribution of seasonal T_{min} values by cluster (April to Oct); **B**) monthly distribution of T_{min} values by cluster; **C**) T_{min} values by cluster before the growing season (January to April) and early (May to July) and late (August to October) in the growing season; and **D**) monthly distribution of T_{min} values by cluster. Colors differentiate between clusters, indicating seasons with different heat levels: Cluster 1, postveraison heat; Cluster 2, preveraison heat; Cluster 3, cool seasons. When present, points and error bars represent means $\pm$ standard error (SE) by cluster. Ribbons represent SE values in line charts.

Supplemental Figure 12 Changes in temperature by cluster according to phenological intervals (BB, budbreak; FL, flowering; VER, veraison; H, harvest). **A**) Minimum temperature (T_{min}), **B**) average temperature (T_{avg}), and **C**) maximum temperature (T_{max}). Points and ribbons represent means $\pm$ standard error. Colors differentiate between clusters, indicating seasons with different heat levels: Cluster 1, postveraison heat; Cluster 2, preveraison heat; Cluster 3, cool seasons.

Supplemental Figure 13 Rainfall features summarized by weather cluster. **A**) Distribution of seasonal rainfall by cluster (April to October); **B**) monthly distribution of rainfall by cluster; **C**) rainfall by cluster before the growing season (January to April) and early (May to July) and late (August to October) in the growing season; and **D**) rainfall by cluster according to phenological intervals (BB, budbreak; FL, flowering; VER, veraison; H, harvest). Colors differentiate between clusters, indicating seasons with different heat levels: Cluster 1, postveraison heat; Cluster 2, preveraison heat; Cluster 3, cool seasons. When present, points and error bars represent means $\pm$ standard error (SE) by cluster. Ribbons represent SE values in line charts.

Supplemental Figure 14 Exploratory analysis of harvest date (**A**) and yield (**C**) and cluster classification of seasons at the five sites (S1 to S5). In **A** and **C**, points represent data available at each site, with the number of observations by site annotated above the data distributions. In **B**, colors differentiate between clusters, indicating seasons with different heat levels: Cluster 1, postveraison heat; Cluster 2, preveraison heat; Cluster 3, cool seasons. DOY, day of year.

Supplemental Figure 15 Effect of weather clusters on Cabernet Sauvignon yield, according to vineyard age. The same analysis was repeated after classifying vineyard blocks as young (3 to 5 yr, left graph) or mature (>5 yr, right graph). Points represent average yield for individual blocks summarized by cluster. Colors differentiate between clusters: Cluster 1 (C1; POST-V, postveraison heat), Cluster 2 (C2; PRE-V, preveraison heat), and Cluster 3 (C3; Cool, cool season). Gray lines connect averages for the same vineyard block belonging to different clusters. Density plots represent distributions by cluster for each age group. Labels report cluster means and 95% confidence intervals (CI) by group, calculated using a linear mixed model where clusters were the fixed effect and vineyard blocks were the random effect on the model intercept. Different letters denote statistical differences between clusters for each age group according to Tukey's adjusted multiple comparison post-hoc test with $\alpha \leq 0.05$.

Supplemental Figure 16 Exploratory analysis of grape analytes measured at five different sites. **A** to **L** represent single fruit composition parameters and points represent data available for each site (S1 to S5). The number of observations by site is annotated on top of data distributions in the first plot only. IBMP, 3-isobutyl-2-methoxypyrazine; TSS, total soluble solids; YAN, yeast assimilable nitrogen.

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Data Availability

The data underlying this study were deposited into Figshare at <https://doi.org/10.6084/m9.figshare.28486232>.

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